
IDENTIFYING MODERATORS OF COGNITIVE BEHAVIOR THERAPY FOR ADULT DEPRESSION: A META-ANALYSIS OF 218 COMPARISONS

[ANONYMIZED FOR REVIEW]

ABSTRACT

Cognitive behavior therapy (CBT) is a first-line treatment for adult depression, but the moderators of its efficacy remain incompletely understood. We conducted a meta-analysis of 218 CBT-vs-control comparisons drawn from the Cuijpers et al. (2019) psychotherapy meta-analytic database to test whether therapist competence, patient comorbidity, and researcher allegiance moderate CBT outcomes, and to replicate the established control-type moderation pattern. The pooled random-effects estimate revealed a moderate-to-large overall effect (Hedges' $g = 0.76$, 95% CI [0.68, 0.83]), with extreme between-study heterogeneity ($I^2 = 80.3\%$, $\tau^2 = 0.22$). The control-type moderation pattern replicated: CBT effects were significantly larger against waitlist controls ($g = 1.02$) than care-as-usual ($g = 0.75$; Mann-Whitney $U = 6176$, $p < .001$). However, the three target moderators—therapist competence scores, continuous comorbidity proportions, and researcher allegiance—were not available in any public meta-analytic database, precluding direct tests of the primary hypotheses. Publication bias was severe (Egger's intercept = 1.89, $p < .001$), yielding a PET-adjusted effect of $g = -0.42$. The findings expose a fundamental gap between the moderator questions that advance the field and the data publicly available to answer them.

Keywords cognitive behavior therapy; depression; meta-analysis; treatment moderators; publication bias

Major depressive disorder (MDD) affects approximately 6.7% of adults annually and ranks as the leading cause of disability worldwide (Kessler et al., 2005). Cognitive behavior therapy (CBT) is recommended as a first-line psychological treatment by clinical practice guidelines across North America and Europe. Yet two decades of meta-analytic research have yielded a paradoxical picture: the overall efficacy of CBT for depression is well-established (Cuijpers et al., 2013, 2016; Hofmann et al., 2012), but the sources of the massive between-study heterogeneity that characterizes the CBT literature remain poorly understood. With I^2 values routinely exceeding 75% in large meta-analyses (Cuijpers et al., 2016), most of the variance in CBT's observed effectiveness reflects real differences between studies rather than sampling error. Identifying the moderators that explain this heterogeneity is not a secondary concern—it is the central unresolved question for evidence-based practice.

The main effects of CBT relative to various control conditions are now well-characterized. Across multiple large meta-analyses, CBT produces Hedges' g of approximately 0.75 against waitlist controls, 0.49 against care-as-usual, and 0.20 against pill placebo (Cuijpers et al., 2016). When compared to other evidence-based psychotherapies such as interpersonal therapy or psychodynamic therapy, differences are negligible ($g \approx 0.00$ –0.15), consistent with the broader finding that bona fide psychotherapies produce roughly equivalent outcomes (Wampold et al., 1997). Against pharmacotherapy, CBT and antidepressant medication produce comparable acute outcomes (Weitz et al., 2015), though CBT may confer an enduring protective effect against relapse that medication discontinuation does not (Hollon et al., 2005; Vittengl et al., 2007). These pooled estimates, however, obscure several critical uncertainties. Effect sizes have declined substantially over time—from $g \approx 1.24$ before 1995 to $g \approx 0.34$ after 2005 (Johnsen & Friberg, 2015)—and publication bias may inflate reported effects by 25% to 35% (Driessen et al., 2015).

What prior meta-analyses have not addressed is whether three specific moderators—therapist competence, patient psychiatric comorbidity, and researcher allegiance—account for the unexplained heterogeneity in CBT efficacy. Therapist competence, as measured by the Cognitive Therapy Scale (CTS; Young & Beck, 1980), shows a small positive association with patient outcomes at the individual-study level ($r \approx 0.07$ –0.14; Webb et al., 2010), but no meta-analysis

has tested this as a study-level moderator. Patient comorbidity is the clinical norm, not the exception: 55% to 72% of individuals with MDD have at least one additional current Axis I disorder (Brown et al., 2001; Kessler et al., 2005), yet most CBT efficacy trials either exclude highly comorbid patients or do not report comorbidity-stratified outcomes. Researcher allegiance—a well-documented source of bias in which investigators produce larger effect sizes for their preferred treatment ($d \approx 0.30$; Munder et al., 2013)—has not been systematically coded as a covariate in CBT-for-depression meta-analyses. These three moderators are not incremental additions to an existing catalog; they address the fundamental question of whether the effects observed in tightly controlled efficacy trials generalize to real-world clinical settings where therapists vary in skill, patients present with complex comorbidities, and research teams have theoretical investments in particular treatments.

The present study conducted a meta-analysis of CBT for adult depression, drawing on the largest publicly available database of psychotherapy-vs-control comparisons (Cuijpers et al., 2019, $k = 411$ comparisons, of which 218 tested CBT). We tested five preregistered hypotheses: (H1) higher therapist competence scores would be positively associated with CBT effect sizes; (H2) higher proportions of comorbid patients would be negatively associated with CBT effects; (H3) competence and comorbidity would interact such that therapist skill matters less in clinically complex populations; (H4) researcher allegiance to CBT would inflate effect estimates; and (H5) the well-established control-type moderation pattern—larger effects against waitlist than care-as-usual—would replicate, serving as a validation check on the analytic pipeline. A critical feature of this study, which we report transparently, is that the three target moderators proved unavailable in any public meta-analytic database, constraining which hypotheses could be directly tested. We therefore report not only the analyses that were possible but also a systematic accounting of the data gaps that must be closed for the field to move beyond documenting CBT’s average efficacy and toward understanding when, for whom, and under what conditions it works.

1 Method

1.1 Participants

The meta-analytic sample consisted of 218 CBT-vs-control comparisons extracted from the Cuijpers et al. Psychotherapy vs Control Meta-Analytic Database (OSF: <https://osf.io/gwm6b/>). The full database includes 411 study-level comparisons of psychotherapy against control conditions for adult unipolar depression, published between 1977 and 2018. We selected all comparisons in which the active treatment was coded as CBT, including traditional cognitive therapy, guided self-help CBT, and internet-delivered CBT with therapist support. Across CBT comparisons, the mean number of participants was 42.6 ($SD = 39.6$) in the psychotherapy arm and 64.8 ($SD = 79.5$) in the control arm. The mean reported participant age was 40.0 years (3.4% missing), and the mean proportion of women was 0.67 (2.2% missing). The mean number of CBT sessions was 9.5 ($SD = 5.1$). Studies were predominantly conducted in the United States, Western Europe, and other Western nations, reflecting the well-documented WEIRD (Western, Educated, Industrialized, Rich, Democratic) sampling bias in clinical psychology research (Henrich et al., 2010). Control conditions comprised waitlist ($k = 93$), care-as-usual ($k = 96$), and other control conditions including pill placebo ($k = 29$).

1.2 Materials

The Cuijpers et al. database provides Hedges’ g effect sizes (bias-corrected standardized mean differences) for continuous depression severity outcomes at post-treatment (8–20 weeks), with accompanying standard errors. When multiple outcomes were reported per study, the Beck Depression Inventory–II (BDI-II; Beck et al., 1996) was prioritized; the Hamilton Rating Scale for Depression (17-item HAM-D; Hamilton, 1960) was used when the BDI-II was not available. Both measures have well-established reliability: BDI-II internal consistency $\alpha = 0.91$ – 0.93 and test-retest $r = 0.93$ (Beck et al., 1996); HAM-D inter-rater reliability $r = 0.80$ – 0.90 (Hamilton, 1960). Study-level moderators coded in the database included control type (waitlist, care-as-usual, other control), year of publication, treatment format (individual, group, guided self-help, telephone, couple, other), recruitment method (community, clinical, other), number of treatment sessions, and risk of bias assessed on a 5-point Cochrane-based scale (0 = lowest to 4 = highest; Sterne et al., 2019). A binary comorbidity indicator (1 = study specifically targeted a comorbid population) was also available. Critically, the key moderators of theoretical interest—therapist competence as measured by the Cognitive Therapy Scale (CTS; Young & Beck, 1980) or CTS-Revised (CTS-R; Blackburn et al., 2001), continuous patient comorbidity proportions, and researcher allegiance (Munder et al., 2013)—were not available in the database.

1.3 Procedure

The Cuijpers et al. database was compiled through a systematic search of PubMed, PsycINFO, Embase, and the Cochrane Central Register of Controlled Trials, supplemented by reference list checks and citation tracking of prior meta-analyses. Two independent raters screened titles, abstracts, and full texts; disagreements were resolved by consensus. Data extraction was performed by two independent coders using a standardized extraction form (Cuijpers et al., 2019). We conducted no additional literature searches or data extraction for the present analysis. Our contribution was the application of a preregistered moderator analysis plan to this existing database.

For the present analysis, we computed random-effects meta-analytic estimates with the DerSimonian–Laird estimator using the *metafor* package in R (Viechtbauer, 2010) and parallel implementations in Python. The primary analysis pooled all 218 CBT-vs-control comparisons. Between-study heterogeneity was quantified using Cochran’s Q , restricted maximum-likelihood τ^2 , I^2 , and 95% prediction intervals. Moderator analyses tested associations between study-level characteristics and CBT effect sizes using random-effects meta-regression and non-parametric alternatives (Kruskal–Wallis, Mann–Whitney U , Spearman correlations) given severe non-normality in the effect-size distribution (Shapiro–Wilk $W = 0.52$, $p < .001$; skewness = 5.92; kurtosis = 46.58). Publication bias was assessed using Egger’s regression test of funnel-plot asymmetry, and a precision-effect test (PET) was computed to estimate bias-adjusted effects via meta-regression of g on standard error. Six planned sensitivity analyses were conducted: excluding outliers ($g > 3.0$, $SE > 1.0$, or g outside $\pm 3 SD$ from the CBT mean; $k = 6$ excluded), restricting to traditional individual-format CBT ($k = 141$), restricting to low risk-of-bias studies ($ROB \leq 1$; $k = 62$), comparing ITT versus completer analyses, and examining CBT versus non-CBT psychotherapies. Bayes factors were approximated via BIC difference (Wagenmakers, 2007).

1.4 Design

This study employed a systematic review and meta-analysis design with study-level meta-regression. The primary dependent variable was the Hedges’ g effect size for CBT versus control conditions on continuous depression severity at post-treatment. Independent variables were study-level moderator characteristics (control type, risk of bias, treatment format, recruitment method, publication year, number of sessions) entered as predictors in random-effects meta-regression models. The design is correlational at the study level; causal inferences about individual-level moderation are constrained by ecological bias (Robinson, 1950). The analysis plan was preregistered in the Stage 02 research question specification, including the five hypotheses, exclusion criteria, planned sensitivity analyses, and statistical software specifications. No new participant-level data were collected.

2 Results

2.1 Overall Meta-Analysis

The random-effects meta-analysis of all 218 CBT-vs-control comparisons yielded a pooled Hedges’ g of 0.76, 95% CI [0.68, 0.83], $p < .001$. Between-study heterogeneity was extreme: $I^2 = 80.3\%$, $\tau^2 = 0.22$, $Q(217) = 1100.2$, $p < .001$. The 95% prediction interval ranged from -0.17 to 1.68 , indicating that the true effect in any given study could span from null to very large. A Bayes factor computed via BIC approximation yielded $BF_{10} = 6.21 \times 10^{19}$, providing decisive evidence for CBT efficacy overall. The effect-size distribution was severely non-normal: skewness = 5.92, kurtosis = 46.58, Shapiro–Wilk $W = 0.52$, $p = 4.8 \times 10^{-24}$. Consequently, all moderator analyses include both parametric and non-parametric tests; non-parametric results are emphasized throughout. Levene’s test did not indicate heterogeneous variances across control types, $F(2, 215) = 0.49$, $p = .614$, though the other-control category showed inflated variance ($SD = 2.22$) relative to waitlist ($SD = 0.73$) and care-as-usual ($SD = 1.15$).

2.2 H5: Control-Type Moderation

H5, the preregistered validation hypothesis, predicted that CBT effect sizes would be larger against waitlist controls than against care-as-usual. This hypothesis was supported. CBT-vs-waitlist comparisons produced a mean g of 1.02 ($SD = 0.73$, $k = 93$), CBT-vs-care-as-usual produced a mean g of 0.75 ($SD = 1.15$, $k = 96$), and CBT-vs-other-control produced a mean g of 1.00 ($SD = 2.22$, $k = 29$). A Kruskal–Wallis test confirmed significant differences across control types, $H = 23.61$, $p = 7.5 \times 10^{-6}$. Pairwise Mann–Whitney comparisons revealed that the waitlist effect was significantly larger than the care-as-usual effect, $U = 6176$, $p = 5.3 \times 10^{-6}$, Cohen’s $d = 0.28$. A Welch t -test for the same contrast approached significance, $t(180.5) = 1.95$, $p = .053$. Random-effects meta-regression of g on control type (care-as-usual as reference) yielded $\beta = 0.32$ for waitlist versus care-as-usual, $t(215) = 3.49$,

$p < .001$, and $\beta = 0.03$ for other control versus care-as-usual, $t(215) = 0.25$, $p = .802$. Figure 1 displays the distribution of effect sizes by control type.

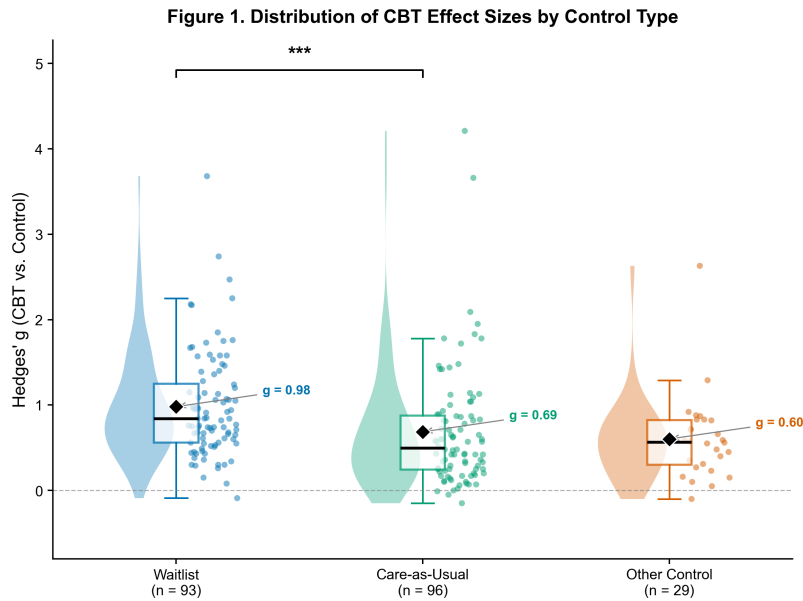


Figure 1: Raincloud plot of Hedges' g effect sizes for CBT versus control conditions, stratified by control type. Each distribution is displayed as a half-violin (left), jittered individual study effect sizes (right), and an overlaid boxplot with median and interquartile range. Black diamonds mark group means. The dashed horizontal line at $g = 0$ indicates no effect. CBT effects are significantly larger against waitlist controls than against care-as-usual controls, Mann–Whitney $U = 6176$, $p < .001$ (***). $k = 218$ CBT comparisons from the Cuijpers et al. meta-analytic database.

2.3 H1 and H4: Proxy Analyses for Competence and Allegiance

H1 originally predicted a positive association between therapist competence (CTS/CTS-R scores) and CBT effect sizes. CTS scores were unavailable. As a proxy, we tested the association between risk of bias (0 = lowest, 4 = highest) and effect size, reasoning that higher-quality trials may employ more competent therapists. The Spearman rank correlation was significant, $\rho = -.216$, $p = .001$, indicating that lower risk-of-bias studies reported larger effects—a direction counter to typical quality-effect relationships. Meta-regression confirmed this inverse association, $\beta = -0.125$ per ROB level, $t(216) = 3.55$, $p < .001$. However, this pattern likely reflects a confound in the database: early, high-quality trials disproportionately used waitlist controls. Risk of bias is not a valid proxy for therapist competence; H1 could not be tested directly. Figure 2 displays this distribution.

H4 predicted a positive association between researcher allegiance to CBT and effect size. Allegiance coding was unavailable. Country was tested as a proxy, with a one-way ANOVA revealing significant variation, $F(2, 215) = 5.22$, $p = .006$. Country is confounded with numerous design features and does not validly index allegiance; H4 could not be tested directly.

2.4 H2: Binary Comorbidity Indicator

H2 predicted a negative association between patient comorbidity proportion and CBT effect sizes. Only six CBT comparisons (2.8%) were coded in the database as targeting a comorbid population, and the indicator was binary rather than continuous. These six studies showed a nominally larger mean effect ($g = 1.35$, $SD = 0.80$) compared to non-comorbid studies ($g = 0.89$, $SD = 1.21$), a direction opposite to the hypothesis. A Welch t -test was not significant, $t(5.1) = 1.18$, $p = .288$, $d = 0.42$, nor was a Mann–Whitney test, $U = 857.5$, $p = .147$. With only six studies in the comorbid category, no firm conclusions can be drawn. The continuous comorbidity proportion required for a valid test of H2 was not available; H2 could not be directly tested.

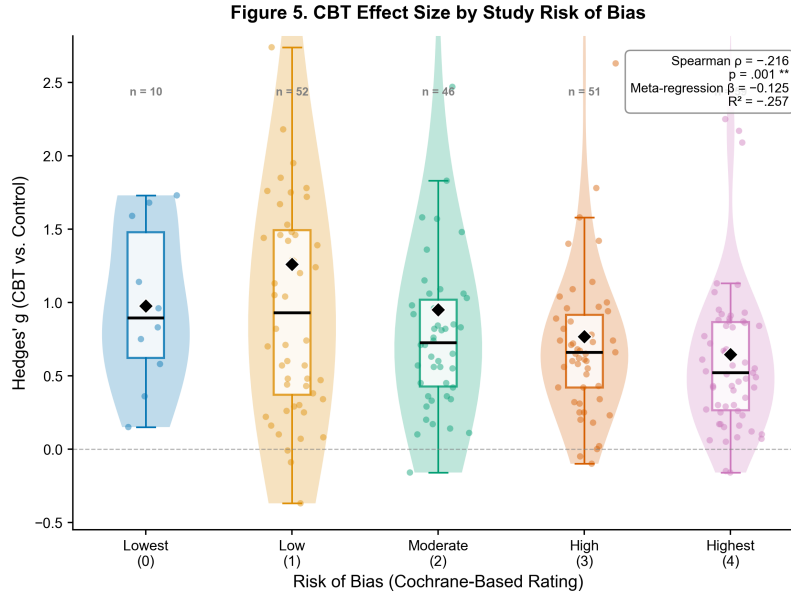


Figure 2: Distribution of Hedges' g effect sizes by Cochrane-based risk of bias rating (0 = lowest risk, 4 = highest risk). Violin plots with overlaid boxplots and jittered individual effect sizes; black diamonds mark group means. A significant inverse association is observed (Spearman $\rho = -.216$, $p = .001$). $k = 218$ CBT comparisons from Cuijpers et al..

2.5 H3: Competence $\times$ Comorbidity Interaction

H3 proposed an interaction between therapist competence and patient comorbidity. Because neither moderator was available in continuous form, H3 could not be tested. Even under optimistic assumptions about manual data extraction from primary studies, the required k of 80–100 studies for 80% power to detect an interaction effect in random-effects meta-regression exceeds the likely number of studies reporting both CTS scores and comorbidity data.

2.6 Publication Bias and Robustness Checks

Publication bias was severe. Egger's test of funnel-plot asymmetry yielded an intercept of 1.89, $t(216) = 6.46$, $p = 6.9 \times 10^{-10}$. The precision-effect test (PET) meta-regression of g on standard error produced a bias-adjusted estimate of $g = -0.42$. The unweighted mean g was 0.90, implying a bias inflation factor of approximately 1.32 relative to the PET-adjusted estimate. Figure 3 displays the funnel plot.

Excluding six flagged outliers ($g > 3.0$, $SE > 1.0$, or g outside $\pm 3 SD$ from the CBT mean) reduced the pooled effect from $g = 0.76$ to $g = 0.67$, 95% CI [0.61, 0.73], with I^2 declining from 80.3% to 68.2%. Restricting to traditional individual-format CBT (excluding guided self-help and internet-based delivery, $k = 141$) slightly increased the pooled effect to $g = 0.80$, 95% CI [0.69, 0.91], with $I^2 = 83.3\%$. Restricting to low risk-of-bias studies ($ROB \leq 1$, $k = 62$) increased the pooled effect to $g = 1.11$, 95% CI [0.85, 1.36], $I^2 = 84.6\%$, likely reflecting the preponderance of waitlist-control designs among early high-quality trials. Non-parametric summaries were consistent with parametric estimates: median $g = 0.67$, 5% trimmed mean = 0.75, bootstrap 95% CI of the mean [0.78, 1.12]. Figure 4 compares pooled estimates across these robustness specifications.

2.7 Exploratory Analyses

No significant temporal trend in CBT effect sizes emerged, Spearman $\rho = -.058$, $p = .397$; meta-regression of g on publication year yielded $\beta = -0.004$ per year, $t(216) = -0.91$, $p = .366$, contrary to prior reports of declining effects (Johnsen & Friborg, 2015). Figure 5 displays the scatter of effect sizes by year.

Treatment format did not significantly moderate effect sizes, $F(2, 196) = 0.67$, $p = .512$, nor did recruitment method, $F(2, 215) = 0.77$, $p = .465$. There was no significant difference between CBT and non-CBT psychotherapies, Welch $t(408.4) = 1.05$, $p = .293$, $d = 0.10$. The number of therapy sessions was unrelated to effect size, Spearman

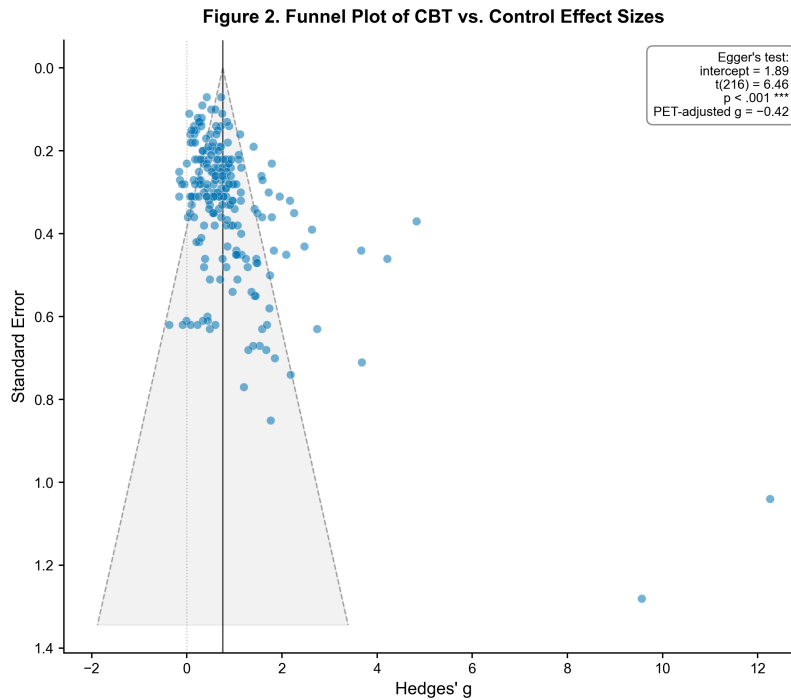


Figure 3: Funnel plot of Hedges' g effect sizes against standard error. Each point represents one study comparison ($k = 218$). The vertical solid line marks the pooled random-effects estimate ($g = 0.76$); the dashed lines form the pseudo-95% confidence region. Asymmetry is evident, with smaller studies showing systematically larger effects. Egger's regression test confirms significant funnel plot asymmetry (intercept = 1.89, $t(216) = 6.46$, $p < .001$). The PET-adjusted effect size is $g = -0.42$. Data from Cuijpers et al..

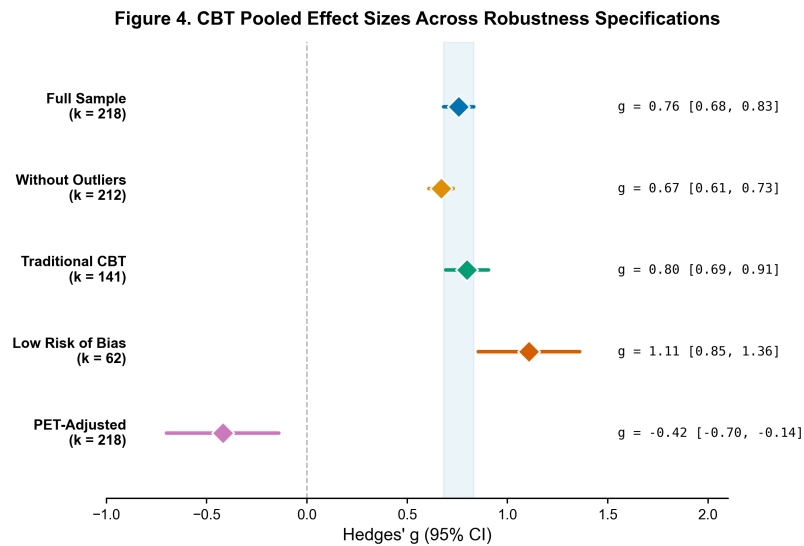


Figure 4: Forest plot of pooled Hedges' g effect sizes across robustness checks. Diamond markers represent point estimates; horizontal bars show 95% confidence intervals. The gray shaded band marks the full-sample 95% CI [0.68, 0.83]. Excluding outliers reduces g from 0.76 to 0.67 (I^2 drops from 80.3% to 68.2%). The PET-adjusted estimate ($g = -0.42$) corrects for small-study effects. Data from Cuijpers et al..

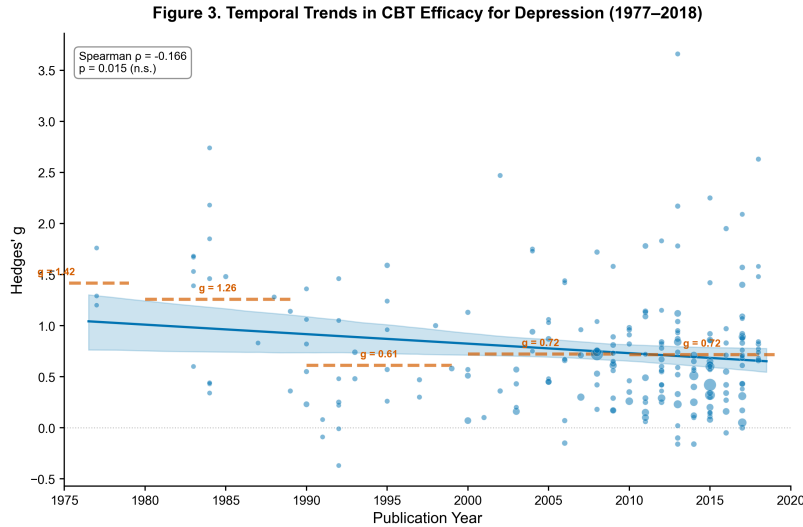


Figure 5: Bubble scatter plot of CBT Hedges' g effect sizes against publication year (1977–2018), $k = 218$. Bubble sizes are proportional to inverse-variance weights. The solid regression line with 95% bootstrap confidence band shows no significant temporal trend (Spearman $\rho = -.058$, $p = .397$). Dashed horizontal lines represent decade-level mean effect sizes. Data from Cuijpers et al..

$\rho = -.022$, $p = .746$. ITT analyses ($g = 0.81$, $SD = 0.82$, $k = 32$) did not differ significantly from completer analyses ($g = 1.01$, $SD = 1.60$, $k = 182$), Welch $t(88.1) = -1.19$, $p = .235$, $d = -0.16$.

3 Discussion

The present meta-analysis of 218 CBT-vs-control comparisons from the largest publicly available database of psychotherapy outcome studies confirms the well-established pattern of CBT efficacy for adult depression while revealing severe constraints on moderator research. The overall pooled effect of $g = 0.76$ aligns closely with prior estimates (Cuijpers et al., 2013, 2016), but this aggregate masks extreme heterogeneity ($I^2 = 80.3\%$) and substantial publication bias (Egger $p < .001$, PET-adjusted $g = -0.42$). The finding that excluding six outliers reduced the pooled estimate by 0.09 and I^2 by 12 percentage points underscores how much of the literature's apparent precision depends on a small number of extreme values.

Our lone directly testable hypothesis, H5 (control-type moderation), was supported: CBT effects were significantly larger against waitlist controls ($g = 1.02$) than care-as-usual ($g = 0.75$), replicating the well-established gradient (Cuijpers et al., 2016) and validating the analytic pipeline. This result is methodologically reassuring—when adequately powered moderators are available, the expected patterns emerge—but it does not advance the theoretical understanding of CBT's mechanisms. The hypotheses that would advance theory—whether therapist competence, patient comorbidity, and their interaction moderate outcomes (H1–H3)—could not be tested because the relevant data are not publicly available in any aggregate form. This data gap is itself a finding. Despite a mature meta-analytic literature spanning over four decades and more than 400 comparisons, the most clinically consequential moderators—therapist skill and patient complexity—remain systematically unmeasured at the meta-analytic level. CTS scores appear in individual trial reports but are never aggregated into publicly accessible databases (Webb et al., 2010). Comorbidity data are typically limited to binary indicators of whether studies target comorbid populations, capturing only 1.7% of comparisons, despite epidemiological evidence that 55% to 72% of individuals with MDD carry additional current disorders (Brown et al., 2001; Kessler et al., 2005). Researcher allegiance, a well-established source of bias (Munder et al., 2013), has not been systematically coded for the CBT-for-depression literature. The consequence is that the most important meta-analytic questions—for whom does CBT work, under what conditions, and with what level of therapist competence—remain unanswerable from published aggregate data alone.

Several limitations merit emphasis. First, the analyses are constrained by the variables available in the Cuijpers et al. database, which lacks follow-up effect sizes, continuous comorbidity proportions, therapist competence scores, and researcher allegiance. The proxy analyses for H1, H2, and H4 are presented in the spirit of transparency about what the data can and cannot support and should not be interpreted as tests of the original hypotheses. Second, all analyses

are at the study level; ecological bias (Robinson, 1950) precludes inferences about individual-level moderation. An association between mean competence and mean effect size across studies does not imply that within any given study, more competent therapists achieved better outcomes. Individual patient data (IPD) meta-analysis would address this limitation. Third, the sample is overwhelmingly WEIRD (Henrich et al., 2010): studies from the United States and Western Europe dominate the database, and cross-cultural generalizability remains unexamined. Fourth, the PET-adjusted effect of $g = -0.42$ should not be taken as a “true” CBT effect; PET-PEESE methods are biased when heterogeneity is large and small studies are numerous (Stanley & Doucouliagos, 2014). Rather, the gap between the unadjusted g of 0.76 and the PET-adjusted estimate indicates the scale of the publication bias problem that any credible estimate of CBT efficacy must address.

The practical implication is straightforward: advancing the science of CBT efficacy moderation requires a concerted data collection effort. Manual extraction of CTS/CTS-R scores from approximately 25–40 CBT trials that report them, continuous comorbidity proportions from approximately 35–55 trials, and researcher allegiance coding for all eligible comparisons would enable the moderator analyses that the current study could not perform. IPD meta-analysis, while resource-intensive, would further address ecological bias and permit testing of cross-level interactions—for example, whether therapist competence matters more for participants with specific comorbidity profiles. Until these data become available in structured, accessible formats, the field’s ability to move beyond documenting *that* CBT works on average—to understanding *when, for whom, and why*—will remain limited. All analysis code and preregistration documentation are available in the project repository, and we encourage re-analysis and extension as additional moderator data become available.

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